Bo Shang, Ph.D.

Research Scientist | Robotics & Multimodal Perception

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Robotics and AI researcher focused on reliable perception and field robotics for transportation and infrastructure. Combines multimodal sensing, computer vision, and experimental evaluation with working robotic systems. Research spans roadside LiDAR, vulnerable-road-user safety, robotic inspection, and vision-based UAV control.

Selected Research Experience

University of Maryland | Jul 2026 - Present

Postdoctoral Associate, Civil and Environmental Engineering / Maryland Transportation Institute

- Develop and integrate hardware and software for two sponsored transportation research projects under Professor Xianfeng Yang; support system testing, field data collection, and technical deliverables.
- Connect research requirements with working sensing systems through troubleshooting, documentation, and review of data quality.

The City College of New York, CUNY | Jul 2025 - Jun 2026

Postdoctoral Scholar, AI & Mobility Research Lab

- Developed roadside LiDAR-camera research workflows covering capture, synchronization, calibration, reconstruction, annotation, and road-user detection for transportation safety.
- Investigated cross-configuration generalization and evaluated models across sites with different sensor beam counts, frame rates, and mounting geometries; published first-author work on LiDAR sensing and vulnerable-road-user detection.

The City College of New York, CUNY | Dec 2022 - Dec 2024

Postdoctoral Researcher, Robotics Lab

- Developed CNN-based structural-defect analysis and robotic inspection workflows linking sensing, cloud inference, 3D reconstruction, and engineering review.
- Built an iOS interface for a climbing inspection robot and integrated human review into inspection-data workflows; co-authored research in IEEE Transactions on Automation Science and Engineering.

Missouri University of Science and Technology | Jan 2020 - Nov 2022

Postdoctoral Fellow

- Developed UAV guidance and clamping-robot components for bridge inspection, combining embedded perception, control, and system testing.
- Integrated ROS, NVIDIA Jetson, Raspberry Pi, and inter-robot communication; co-inventor on U.S. Patent 12,296,994 B2, granted in 2025.

Education

Northeastern University, China | 2013 - 2020

Ph.D., Pattern Recognition and Intelligent Systems

University of California, Merced, USA | 2015 - 2017

Exchange doctoral research

Selected Research

Reliable roadside perception | Nov 2024 - Jun 2026

- Investigated how reduced LiDAR beam coverage affects vulnerable-road-user detection, using comparative model evaluation to examine sensing limits.
- Developed a roadside safety-auditing framework linking trajectories, iterative human quality assurance, and near-miss analysis; a heavy-vehicle/bicycle case distinguished lateral intrusion from longitudinal collision risk.

Vision-based control for field robotics | 2015 - 2019

- Developed and evaluated vision-based guidance and fractional-order control methods for quadrotor UAVs, including precision hovering with longer sampling periods; published first-author work in Nonlinear Dynamics.

Selected Publications

Roadside LiDAR for Cooperative Safety Auditing at Urban Intersections: Toward Auditable V2X Infrastructure Intelligence. Bo Shang, Yiqiao Li. *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) Workshops, pp. 757-765*, 2026. [Conference paper] arXiv PDF Record

Drive-by Urban Freight Sensing: Large-Scale Fine-Grained Truck Classification Using Google Street View Imagery and Deep Learning. Bo Shang, Yiqiao Li. *IEEE Open Journal of Intelligent Transportation Systems, vol. 7, pp. 2327-2347*, 2026. [Journal article] DOI PDF Record

Robotic Inspection and Data Analytics to Localize and Visualize the Structural Defects of Concrete Infrastructure. Jinglun Feng, Bo Shang, Ejup Hoxha, César Hernández, Yang He, Weihan Wang, Jizhong Xiao. *IEEE Transactions on Automation Science and Engineering, vol. 22, pp. 22324-22336*, 2025. [Journal article] DOI Record

Technical Expertise

Research and evaluation: Cross-site evaluation, dataset and configuration versioning, annotation quality assurance, model evaluation, failure analysis, and reproducible research workflows.

Robotics and sensing: LiDAR, RGB/RGB-D cameras, sensor synchronization and calibration, ROS / ROS 2, NVIDIA Jetson, embedded Linux, visual SLAM, and UAV perception and control.

Software and machine learning: Python, C++, PyTorch, TensorFlow, OpenCV, AWS, Docker, Git; computer vision, 3D reconstruction, data pipelines, and cloud and edge inference.